The first right aroow in the above equations means λ and z are the eigenvalues and the eigenvectors for $\boldsymbol{A}\boldsymbol{A}^T$. So the $\boldsymbol{z}$ we want to derive from the formula should be the eigenvectors corresponding to the eigenvalues λ sorted from largest to smallest.

F Reconstruction Error of SAE

We also experiment with the relationship between reconstruction error and the number of selected features in `Gemma-2-9b-it`. The result is shown in Figure 6, when $k = 10$, the reconstruction percentage exceeds 90%. This proves that our sparsity assumption is reasonable to some extent.

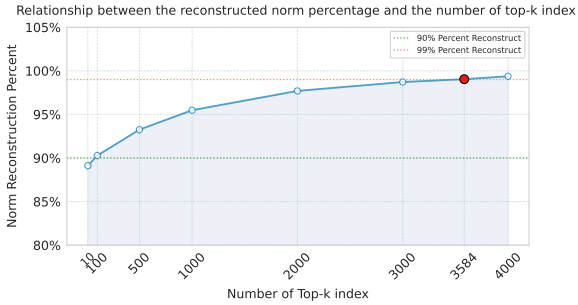


Figure 6: Relationship between reconstruction error and the number of selected features(Gemma-2-9b-it)

G Approximate Orthogonality of SAE feature vectors

In order to explore the geometric relationship between top-k feature vectors of SAE, we calculated the cosine similarity matrix. A similarity of 0 indicates that the two are orthogonal. We utilize `llama_scope_r1_distill(l15r_800m_openr1_math)` as an example, the result is shown as Figure 7, we can find that for top-150 feature vectors, they are almost pairwise orthogonal.

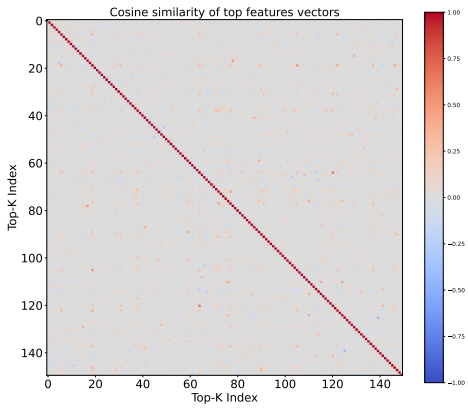


Figure 7: Cosine Similarity of Top-k feature vectors

H Generating Verbal CoT and Symbolic CoT

We give a prompt example to instruct GPT-4.1 to generate Verbal CoT and Symbolic CoT that were used to extract features in Figure 8 and Figure 9.

System: You are a helpful assistant who can follow this command. COMMAND: Give the general idea of the following problem.m Only state the steps, including reasoning words such as 'therefore', 'but', 'however', 'wait' as much as possible, but do not include any math-related symbols, formulas and calculations.

User: The function f satisfies the functional equation

$$f(x) + f(y) = f(x + y) - xy - 1$$

for all real numbers x and y. If $f(1) = 1$, then find all integers n such that $f(n) = n$. Enter all such integers, separated by commas.

System: Here's a natural English explanation without mathematical symbols:
Step 1: Understanding the Problem We're dealing with a special function that follows a specific rule: when you take any two numbers, apply the function to each separately and add those results, it's equivalent to applying the function to their sum first, then subtracting their product, and finally subtracting one. We're also told that when we input one into this function, we get one as output. Our goal is to find all whole numbers where the function's output equals the input itself.
Step 2: Making an Educated Guess About the Function Problems like this typically involve polynomial functions. Since we see the product of two variables appearing in the rule, this strongly suggests our function is quadratic - meaning it likely involves terms where the variable is squared, along with other components.
Step 3: Setting Up the Relationship We'll propose a general form for our quadratic function with three unknown coefficients. By plugging this form into the original rule and carefully expanding both sides, we can rearrange terms to see how different components relate to each other.
Step 4: Comparing Components After expansion, we'll systematically match up similar terms from both sides of the equation - those involving squares of variables, products of variables, single variables, and constant terms. This will give us equations to determine our unknown coefficients.
Step 5: Using the Given Information We know specifically what happens when we input the number one into our function. This provides an additional concrete equation that will help pin down the exact values of our coefficients.
Step 6: Solving for the Unknowns With these equations established, we can now solve for all three coefficients. Once we've completely determined our function's exact form, we'll be able to evaluate it for any input.
Step 7: Finding Special Values Finally, we'll look for those particular whole numbers where the function's output is identical to the input itself. This involves setting up and solving another equation based on the complete function we've determined.
Conclusion: Through this step-by-step approach, we'll fully characterize the function and identify all integer values that satisfy our special condition where the function returns exactly what was input.

User: `{question}`

Figure 8: The prompt for generating Verbal Process.

System: You are an assistant who can follow this command. COMMAND:Give all the mathematical calculations for the following questions.Do not include any text description and logical derivations.

User: The function f satisfies the functional equation

$$f(x) + f(y) = f(x + y) - xy - 1$$

for all real numbers x and y. If $f(1) = 1$, then find all integers n such that $f(n) = n$. Enter all such integers, separated by commas.

System: 1. **Assume quadratic form:**

$$f(x) = ax^2 + bx + c$$

2. **Substitute into the functional equation:**

$$ax^2 + bx + c + ay^2 + by + c = a(x + y)^2 + b(x + y) + c - xy - 1$$

3. **Expand and simplify:**

$$ax^2 + ay^2 + bx + by + 2c = ax^2 + 2axy + ay^2 + bx + by + c - xy - 1$$

4. **Cancel common terms:**

$$2c = 2axy - xy + c - 1$$

5. **Collect like terms:**

$$(2a - 1)xy + (c - 2c + 1) = 0$$

6. **Set coefficients to zero:**

$$2a - 1 = 0 \quad \Rightarrow \quad a = \frac{1}{2}$$

$$-c + 1 = 0 \quad \Rightarrow \quad c = 1$$

7. **Use $f(1) = 1$:**

$$\frac{1}{2}(1)^2 + b(1) + 1 = 1 \quad \Rightarrow \quad \frac{1}{2} + b + 1 = 1$$

$$b = -\frac{1}{2}$$

8. **Final form of $f(x)$:**

$$f(x) = \frac{1}{2}x^2 - \frac{1}{2}x + 1$$

9. **Solve $f(n) = n$:**

$$\frac{1}{2}n^2 - \frac{1}{2}n + 1 = n$$

$$n^2 - 3n + 2 = 0$$

$$(n - 1)(n - 2) = 0$$

$$n = 1, 2$$

Final Answer:

$$\boxed{1, 2}$$

User: `{question}`

Figure 9: The prompt for generating Symbolic Process.

I Eigenvector Steering Visualization

To explore in more detail how SAE-free works and how its role is similar to SAE-based steering, we performed SAE-free steering on two LLMs with associated SAEs:`DeepSeek-Llama3-8B` and`Gemma-2-9b-it`. The setting is the same as before. We then observe the activation changes on certain SAE features before and after SAE-free steering. Figure 10 and Figure 11 show the activation change of feature 20903 and 4118 for `llama_scope_r1_distill(l15r_800m_openr1_math)`. 4118 represents "mathematical parameters related to dimensions and positioning". After steering, some related words like "lengths" and "height" are significantly activated. Figure 12 and Figure 13 show the activation change of feature 8927 and 13526 for `gemma-scope-9b-it-res(layer_9/width_16k/canonical)`. Both figures also exhibited the fact that some words related to mathematics and reasoning were significantly activated.

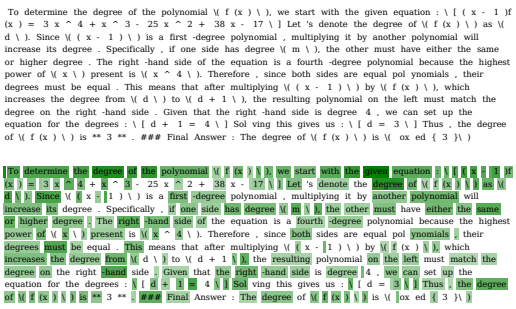


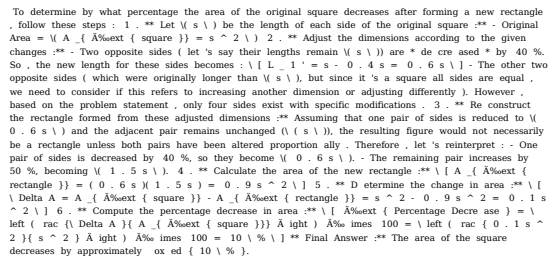
Figure 10: Eigenvector steering visualization for feature idx 20903(structures and elements related to mathematical expressions and equations) for `DeepSeek-Llama3-8B`.

J Explanation of Eigenvector

In order to verify that the obtained eigenvectors have good interpretability, we take the eigenvectors corresponding to the top-50 eigenvalues, perform inner product operations on them and the d_{SAE} feature vectors of SAE to obtain the similarity. A total of $50 * d_{\text{SAE}}$ operations are performed to obtain the top-10 similarities, and the results are shown in Table 4 and Table 5.

K Further Analysis about Feature Selection

To further illustrate that the subtraction method is more advantageous than the addition method, in-



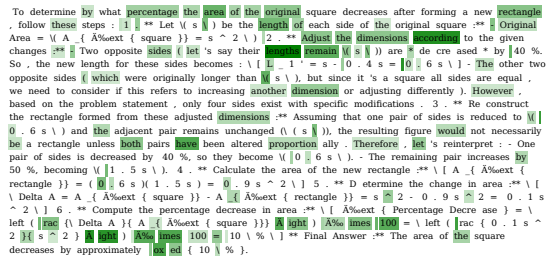
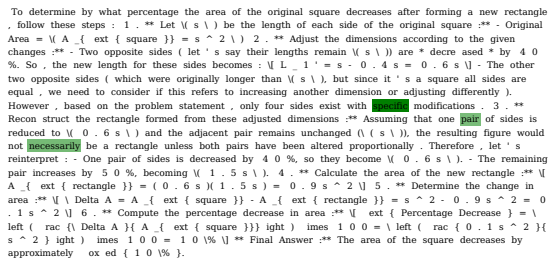


Figure 11: Eigenvector steering visualization for feature idx 4118(specific mathematical parameters related to dimensions and positioning) for `DeepSeek-Llama3-8B`.



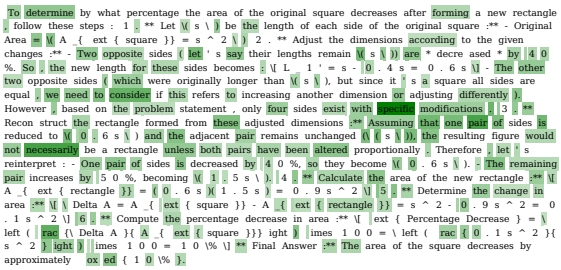


Figure 12: Eigenvector steering visualization for feature idx idx 8927(phrases related to mathematical combinations and choices) for `Gemma-2-9b-it`.